

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Farmers, Researchers, Policymakers, Admin) can securely log in using email/mobile number and password.	High
2	Authorization levels	Farmers can get crop recommendations, Researchers can analyze crop-environment data, Policymakers can access reports, and Admin manages users and datasets.	High
	External interfaces	Integrates with agricultural databases, weather/rainfall datasets, and soil information sources for accurate	High

		recommendations.	
4	Transactions processing	The system accepts user inputs (N, P, K, temperature, humidity, pH, rainfall), processes them using the prediction model, and returns crop recommendations instantly.	High
5	Reporting	Generates crop recommendation reports, environmental suitability analysis, dashboards, and downloadable summaries for users.	Medium
6	Business rules, etc.	Crop recommendations are generated only after all mandatory environmental parameters are entered. The prediction follows the trained machine learning model.	High
7	Compliance to laws or regulations	User data is stored securely and handled according to applicable data privacy regulations.	Medium
8	<i>Other</i>	Provides crop suitability assessment and supports agricultural research through historical data analysis.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	Crop recommendations should be generated quickly after user input.	Response <i>time less than 2 seconds</i>
2	Scalability	The system should support increasing numbers of users and agricultural datasets.	Supports 10,000+ users and large datasets
3	Security & Data Privacy	Protect user information using secure authentication and encrypted storage.	Password encryption, HTTPS, secure database access
4	Reliability & Availability	The system should be available whenever users need it with minimal downtime.	99.9% uptime
5	Usability & Accessibility	Provide a simple, responsive, and user-friendly interface accessible on desktop and mobile devices.	Easy navigation with responsive design
6	<i>Other</i>	Maintainability and portability to support future updates and deployment on different platforms.	Modular code with easy maintenance and cloud deployment support